Multi-Head Plant DiseaseClassification for African Food Security Using MobileNetV3-Small

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Abstract-In Africa, for instance, more than 700 million people rely on food like Cassava and Corn to survive. Yet, food security is always at risk due to diseases that affect these crops, resulting in the loss of at least 40% of total production. Research by Cornell University and the IPCC shows that climate-related variables have already contributed to a 34% decline in agricultural productivity growth in Africa, making it a humanitarian imperative to identify diseases accurately.

The main challenge is inaccurate diagnosis. If a farmer diagnoses a problem incorrectly, for example, **Cassava Mosaic** Disease instead of a nutritional deficiency, crop production can decline by as much as 80-90%. This problem is further exacerbated by "wasteful expenditure," where scarce resources are allocated to inefficient solutions. This paper fills the "information gap" by proposing a multi-head plant and disease classifier based on the MobileNetV3-Small model, selected for its superior performance and efficiency on resource-constrained hardware.

The research focuses on five significant African crops, namely Corn, Cassava, Tomato, Cowpea, and Cocoa. One of the main problems faced in agricultural AI is that the model tends to degrade with real-world variables such as erratic lighting conditions and background noise. This problem was overcome using advanced image processing techniques, ensuring that the final model was able to attain a 99.97% F1 score in 6-class plant classification and 98.49% F1 score in 27-class disease classification (encompassing 21 diseases, 5 healthy states, and out-of-distribution rejection).

The model will be deployed using an offline API, which is significant in that it helps to overcome the digital divide, ensuring that African farmers can attain instant expert-level diagnostic results without access to the internet. This helps to ensure that the 78% of Africans who cannot afford healthy foods can attain a secure nutritional future.

Keywords: MobileNetV3, Crop Disease, Africa Food Security, Deep Learning, Offline AI, Precision Agriculture.

INTRODUCTION

The problem of the degradation of machine learning models when exposed to real-world situations in which lighting and background noise affect the overall performance of the models was addressed in this study by employing advanced image processing techniques. In the final iteration of the model, it achieved a 99.97% F1 score in 6-class plant classification and 98.49% in 27-class disease classification. The API was used to deploy the model in an offline setup. This is important in addressing the problem of disease diagnosis in plants across the African continent. This study proves that deep learning models can be used to address critical ecological and humanitarian issues in Africa. The paradox of African agriculture is staggering: while global farming technology has taken off since 1961, African agriculture productivity growth has lagged with a substantial 34% decline. According to research conducted by Cornell University and the IPCC, this decline has placed a continent of immense potential in a position of nutritional precariousness. In fact, today 78% of Africans, or over one billion people, cannot afford a healthy and nutrient-dense diet. While climate change is a major factor, a more proximate and tractable 'silent killer' is hiding in plain sight: crop disease misdiagnosis.

In Africa, for example, plant diseases and pests are known to cause at least 40% of the overall yield losses. However, the tragedy is not the existence of these pathogens, but the inability to accurately identify them. For small-scale farmers, who are the majority of the producers of the continent's food basket, they rarely get access to professional agronomists. If a farmer incorrectly identifies a pathogen, the consequences are disastrous. For example, if a farmer incorrectly identifies Cassava Mosaic Disease (CMD) as a nutrient deficiency or a different bacterial infection, the annual yield losses could be as high as 80% to 90%.

The "information gap" has made poverty a double-edged sword. It destroys the food supply, driving up the price of healthy staples until they are a luxury. It also causes unproductive spending. A farmer living on less than $2 a day may spend her last dollars on a fungicide to treat a crop that is really infected by a virus or bacteria. Not only will this treatment do no good, but it will also consume the capital needed to replant the crops, leaving the farmer in a cycle of debt.

To stop this vicious cycle, it is imperative that we identify the root of the problem, and that root is the unavailability of a trusted diagnostic tool. The answer, therefore, lies in infusing the brain of an agronomist into the palm of every farmer's hand. For that, this research selected MobileNetV3, a state-of-the-art neural network architecture created by Google. MobileNetV3 is optimized for "performance and efficiency," making it the perfect tool for image classification tasks, especially under limited resources.

With the development of a multi-head plant and disease classifier, this research aims to build a tool that goes beyond merely identifying a diseased leaf. With a multi-head approach, it is possible to first identify the type of crop and, at the same time, identify the type of disease affecting it. The use of MobileNetV3 ensures that it is possible to use TensorFlow Lite, enabling offline use of the tool, as it is likely that the farmers who require it most will be located in areas with limited or no 4G connectivity.

By replacing guesswork with precision, we can safeguard the 40% of our crop yields that are being lost to disease every year. Precision in identification means precision in treatment, resulting in increased crop yields, savings for our farmers, and ultimately, a major impact on our economy. When we address this diagnostic deficit, we are not just saving a crop; we are giving 78% of our population a chance at food that is not only affordable but healthy as well.

WHY THIS MODEL IS DIFFERENT

In reality, it is not only this research that has created such models, and many individuals and organizations have created them, but the models created do not fully digest the intricacy of the problem. Some of the primary issues are the use of lab-grown data for training, models not being able to handle all inputs as plants, the monolithic structure of the classifiers, and the computational costs. First and foremost, when a model is trained on lab-grown data (images), it is more likely for the model to fail when it is tested on field images, and such a fact is also corroborated by the users of such models. Models trained on lab-grown datasets are seen to be "state-of-the-art" on paper, but there is a significant gap in the performance of the models when tested on images from other sources, such as field images. This "bug" can be attributed to the fact that lab images are not subjected to the same lighting as crops are subjected to the sun; lab images are just too clean and do not depict actual crop data since their backgrounds are totally different from the ones found in the farmlands. To solve this, this research ensured that the dataset used has a majority from "the wild" and a small percentage from the lab, and this has been supplemented with various data augmentation methods to make the training process robust.

Moreover, the majority of the existing models have a tendency to misclassify non-plant images as plants since there are a number of model architectures that have failed to incorporate a class of images that are "out-of-distribution" (OOD). In multi-class classification problems using deep learning, the ultimate aim of the final linear layer is to generate logits that are then used by the Softmax activation function to generate confidence scores. Therefore, if there is no class that represents out-of-distribution images, the model, by applying the argmax function, forces itself to classify the input image as a specific plant with a specific disease, even if it is not a plant at all. In accordance with best practices, this research used five different types of plants and added one specific class that represents out-of-distribution images, making it a total of six classes in the plant classification model. The research also used 27 classes in the disease classification model, where 21 represent plant diseases, five represent healthy versions of the plants, and one represents out-of-distribution images.

The other critical issue with the models, which many researchers fail to consider, is the computational efficiency. Many of the current models are based on heavy architectures such as ResNet-101 and InceptionV3, with tens of millions of parameters and therefore are extremely costly. At the same time, the majority of the farmers use small devices. When a model based on heavy architecture is deployed on the devices, the deployment will be extremely costly, or the model will not execute at all since the devices are not capable of executing the model. It will therefore not be clear if the problem has been solved for the end user. As the best option for the case, this research chose the MobileNetV3 architecture. It was developed specifically to bridge the gap between high-performance and computational efficiency. It has over 15 times fewer parameters compared to the regular heavy architecture.

CHAPTER 3: ARCHITECTURAL METHODOLOGY

3.1 Overview of MobileNetV3-Small

The choice of MobileNetV3-Small as a primary feature extractor is based on its distinct development background. Unlike other networks that were designed through manual tuning of various hyperparameters, MobileNetV3-Small was developed through a collaborative process of Hardware-Aware Network Architecture Search and the NetAdapt method, focusing on the optimization of the network for mobile CPU low-power conditions. The architecture of this network can be described as its modularity, based on a refined version of the Inverted Residual Block referred to as MBConv, including squeeze and excitation modules and a hardware-efficient activation function called Hard Swish.

3.2 The Input and Initial Feature Extraction

The architectural pipeline starts with an input tensor

X

X of size $\mathbb{R}^{3\times224\times224}$

. Although the model is technically capable of handling any resolution, the standard resolution of

224×224

224×224 is maintained for efficiency.

3.2.1 The Stem Convolutional Block

The initial block of the network is the Conv2dNormActivation layer. This is not a regular convolutional layer but a sequential container for three different operations: 3x3 Standard Convolution: In this layer, 16 filters with a stride of 2 are used. This reduces the spatial size to half, i.e., 112×112 with a depth of 16. The convolutional operation is defined by the sliding of a kernel K over the input I, where the value of the element in the feature map is given by the dot product:

$$(I * K)(i, j) = \Sigma\square\ \Sigma\square\ I(i + m, j + n) \cdot K(m, n)$$

Batch Normalization (BN): The BN layer is directly added after the convolution layer to stabilize the training process. It normalizes the activations by the mean $(\dot{x} = (x - \mu)/(\sigma))$ of the mini-batch to ensure that the next layer receives a stable distribution. This is essential to prevent the "exploding gradient" issue, which is prevalent in deep networks.

Hard Swish Activation: To introduce non-linearity, the Hard Swish activation is used. The regular Swish activation, given by the expression

$$\text{swish}(x) = x \cdot \text{sigmoid}(x)$$

swish, is computationally intensive for mobile hardware due to the exponential calculation involved in the sigmoid activation. MobileNetV3 replaces the sigmoid activation with the piecewise linear ReLU6 activation:

$$\text{h-swish}(x) = x \cdot \frac{\text{ReLU6}(x + 3)}{6}$$

3.3 The Inverted Residual (MBConv) Structure

The basic block of MobileNetV3-Small is called the Inverted Residual Block. This block is a variation of the MobileNetV2 block, which has been further improved using NAS. Unlike standard residual blocks, such as those in ResNet, this block has a narrow-wide-narrow pattern, which might be counter-intuitive. However, it is efficient for mobile devices.

3.3.1 Pointwise Expansion and Latent Space

Each block consists of a 1×1 pointwise convolution. The idea behind using a 1×1 pointwise convolution at the beginning of each block is to map a low-dimensional space to a high-dimensional space, also called the "latent space." For example, if we are given 16 channels as input, we might use an "expansion ratio" of 3x or 6x to expand it to 72 or 96 channels.

3.3.2 Depthwise Convolution

This expanded convolution is followed by the Depthwise Convolution operation. In standard convolutional layers, the convolutional filter is applied across all channels. In the depthwise convolutional layer, the convolutional filter is applied separately to each channel. Thus, in the depthwise convolutional layer, the convolutional filter size can only be 3x3 or 5x5. This reduces the computational complexity by a factor almost equal to the number of output channels, which makes MobileNet so computationally efficient.

3.3.3 Squeeze-and-Excitation (SE) Modules

Another difference between the MobileNetV3-Small model and the MobileNetV2 model lies in the presence of the Squeeze-and-Excitation module in the residual blocks of the MobileNetV3-Small model. The "Squeeze-and-Excitation" module is essentially a "channel-wise attention" module. In the "Squeeze" operation, the spatial information in the feature maps is squeezed into a single descriptor using the global average pooling operation. In the "Excitation" operation, two fully connected layers with a sigmoid activation function are used to obtain the weights, which are used to scale the feature maps to "pay attention" to the features that are significant (such as the texture of the leaves) while ignoring the features that are not significant (such as the background).

3.3.4 Linear Bottleneck and Projection

The final layer of the block is the projection layer, which is a 1×1 layer. This layer is referred to as the "Linear Bottleneck" because it reduces the high-dimensional space to a smaller number of channels. It has been proven that the non-linear activation functions in these layers cause the loss of too much information, and hence the layer is designed to be linear.

3.4 The Efficient Last Stage

One of the most significant changes in the MobileNetV3 network is the efficiency of the final stages of the network. In the previous MobileNet variants, the heavy
1×1 expansion layer was placed before the pooling layer, which required high computation for the
7×7 feature maps. However, the MobileNetV3-Small has placed the layer after the Adaptive Average Pooling.

The feature maps at the end of the backbone network have a representation as a tensor of size 7×7×96. The Adaptive Average Pooling operation collapses this representation to a vector of size 1×1×576. By expanding this operation from a spatial size of 7×7to a spatial size of 1×1, a substantial latency reduction is achieved without any accuracy degradation.

3.5 Custom Multi-Head Classification Architecture

Although the MobileNetV3-Small model is used as the feature extractor, the classification problem is addressed using a custom Multi-Head classification architecture. This is beneficial because the system is able to tackle two separate classification problems, namely Disease Detection and Plant Identification, using the same set of features.

Architecture Summary: The dual-head architecture consists of:

- Plant Identification Head: 6 total output classes (5 crop species + 1 OOD)
- Disease Classification Head: 27 total output classes (21 diseases + 5 healthy states + 1 OOD)

3.5.1 The Disease Classification Head

The Disease Head is designed for precise feature recognition. It uses the 576 feature extraction from the backbone and applies a sequential process:

1. Flattening: The
2. 1×1×576 tensor is flattened into a 1D vector.
3. Dense Dimensionality Reduction: The 576 features are transformed into 256 units using a Linear layer. This gives the model the ability to capture intricate patterns of diseases.
4. Activation (Hard-Swish): To ensure the backbone is preserved, the Hard-Swish function is used. This ensures the gradient flow is optimized for mobile-centric math.
5. Regularization (Dropout): The dropout is set to 0.3. This implies that 30% of the neurons are randomly disabled during training. This makes the neurons learn more robust features and prevents the model from memorizing the training set.
6. Final Projection: The 256 units are projected into 27 units, representing 21 disease classes, 5 healthy plant classes, and 1 out-of-distribution class.

3.5.2 The Plant Identification Head

Concurrent with the Disease Head is the Plant Species Head. Although it has the same input size (576 features), its activation function is different:

1. ReLU Activation: Unlike the Disease Head, this layer employs the Rectified Linear Unit (ReLU). ReLU is given by the expression
2. f(x)=max(0,x)
3. f(x)=max(0,x). The reason for this is that the more extreme thresholding function of ReLU may actually be preferred in the more distinct categorical task of species identification.
4. Symmetric Regularization: As with the disease head, a 256-unit hidden layer and 0.3 dropout are employed to maintain symmetry in the learning capacity of the two tasks.

5. Final Projection: The 256 units are projected into 6 units, representing 5 crop species and 1 out-of-distribution class.

3.6 Summary of Architectural Flow

The final architecture is one of a highly specialized vision pipeline. The input image, initially 224x224 in size, is processed through 11 Inverted Residual layers with a variety of 3x3 and 5x5 kernels, depthwise separable convolution, and channel attention SE modules. The result is a 576-dimensional vector, which serves as input to two independent neural heads as part of a multi-task learning paradigm. From a computational perspective, the implementation of a single backbone with approximately 2.5 million total parameters also ensures a substantial reduction in memory consumption—over 15 times fewer than the 44.5 million parameters in a standard ResNet-101 architecture. This not only reduces memory requirements by requiring only one network for two tasks but also potentially leads to better generalization in botanical diagnostics through the shared feature representation.

ADVANTAGES OF MULTI-HEAD ARCHITECTURE

The Multi-Task 'Inductive Bias' and Knowledge Synergism

The use of a dual-head network on a single MobileNetV3-Small backbone creates a powerful Inductive Bias in the model's learning process. Inductive bias, as a machine learning term, is defined as a set of assumptions that a machine learning model uses to make predictions on inputs it has never before encountered. By requiring the model to optimize both the disease and plant species loss functions, a 'bottleneck of relevance' is created.

The backbone is forced to learn features that are relevant to both tasks, like leaf venation, color gradients, and textures of lesions. By doing this, a form of regularization is created. A model that is only tasked with the detection of diseases might end up overfitting on the specific color of a fungal spot, whereas a multi-task model will realize that this spot is part of a specific leaf structure.

Prevention of Catastrophic Forgetting

One of the problems that can be encountered when dealing with sequential deep learning models, especially in the context of the problem at hand, is the occurrence of Catastrophic Forgetting, where a model trained for Task A (Plant Identification) completely forgets how to perform when fine-tuned for Task B (Disease Detection). This problem, however, is completely circumvented in the present architecture through the implementation of the multi-head architecture, which achieves Simultaneous Parameter Optimization.

The fact that the parameters of the MobileNetV3-Small backbone are adjusted based on the combined gradients of the performance of both the disease_head and the plant_layer means that the model retains a persistent memory of the performance of both tasks. This architectural decision ensures that the internal representations of the model, i.e., the 576-feature vector, remain relevant for both botanical and pathological classification. From a computational perspective, the implementation of a single backbone also ensures a substantial reduction in memory consumption, as the system does not have to maintain the memory for the 2.5 million parameters of both models.

IV. EXPERIMENTS AND RESULTS

This section discusses the experiment setup that was used to train the proposed architecture of the dual-head plant and plant disease classification approach. It includes a discussion of the dataset, the non-standard approach to loading the dataset due to the original structure, the experiment setup, and the performance analysis.

A. Dataset Description

The model has been trained on a specially curated dataset that consists of around 43,000 images, collected from various public domains related to agricultural images. It appears that the dataset has been designed primarily based on the variations of the PlantVillage dataset structure. It was organized in such a manner that it could cater to a dual classification.

Architecture Summary: The dual-head architecture operates on two classification tasks:

- Plant Identification Head: 6 total classes (5 crop species: Maize, Cocoa, Cowpea, Tomato, Cassava + 1 Out-of-Distribution class)
- Disease Classification Head: 27 total classes (21 distinct diseases + 5 healthy plant states + 1 Out-of-Distribution class)

The dataset includes a total of 5 different plant species (Maize, Cocoa, Cowpea, Tomato, Cassava), for which a further class was created for the Out-of-Distribution (OOD) test during the evaluation stage. The dataset encompasses 21 distinct diseases affecting the five target crops. Combined with 5 healthy plant classes (one per crop species) and 1 out-of-distribution class, this results in 27 total classes for the disease classification head. The hierarchical structure of the data (Plant -> Disease -> Images) required a design tailored for data handling, as discussed in Section IV-B.

B. Data Preparation and Loading Methodology

In many cases, standard frameworks for deep learning, such as torchvision, expect a flat structure (e.g., ./class_A/). Our dataset is actually stored in a hierarchical manner: ./plant_type/disease_label/images.jpg. This prevented us from directly using the ImageFolder class.
In order to accomplish this, the Python os library was used to traverse the directory structure programmatically. This allowed the important metadata associated with each of the paths of the images used in this problem set to be leveraged. This metadata was then compiled in the form of a pandas DataFrame. This was used as the overall index in a custom Dataset class that leveraged PyTorch.
A dedicated out-of-distribution (OOD) class was created using 1,200 images of non-plant objects (soil, sky, tools, hands) and non-target plants. These were included in both training (15% of OOD samples) and testing splits to teach the model to reject invalid inputs.
Such a methodology, while more complicated than the basic approach, ensured that the training batch appropriately linked one image input and just two unique target labels (crop type and disease type), which are very important for training the dual head model.

C. Training Configuration

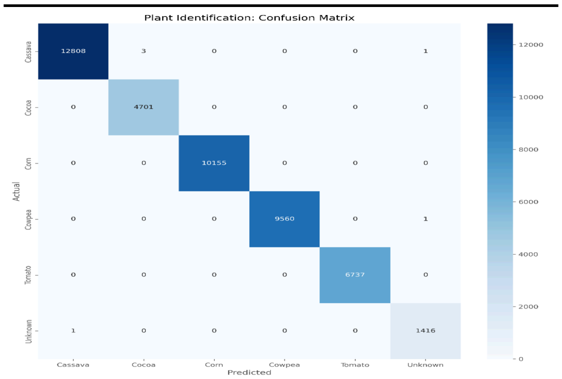
The data was trained with the standard 80/10/10 split for the data in each stage, making sure not to overlap images in different stages. Normal augmentations for the data, like random flip, rotation, and color jittering, were performed using the data.transform consisting of torchvision.transforms protocols, including random horizontal/vertical flips (±30° rotation), color jitter (brightness=0.2, contrast=0.2, saturation=0.2), and normalization. Optimization for the network occurred over normal Adam parameters.

D. Performance Metrics and Analysis

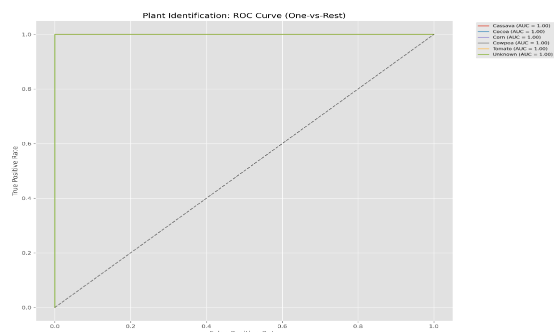
The model was evaluated against three critical parameters: Overall Accuracy, Macro F1-Score (considering the possible class imbalance), and the ROC Area Under the Curve metric, specifically by employing the One-vs-Rest (OvR) Strategy for multi-class evaluation.
1. Plant Classification Head Performance

The first head, responsible for identifying the specific plant species (6 classes: 5 crops + 1 OOD), achieved near-perfect performance:

Metric	Score
Overall Accuracy	99.99%
Macro F1-Score	0.9997
ROC AUC (OvR)	1.0000



Plant Head Confusion Matrix



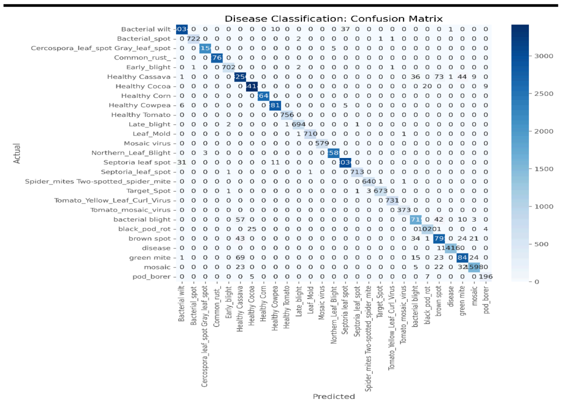
Plant Head ROC curve aand ROC AUC scores
Since the perfect score has been achieved in the AUC test, it shows the model is highly effective in separating the 6 different classes (5 plant species + OOD) with no overlap in the boundaries of classification.

2. Disease Classification Head Performance

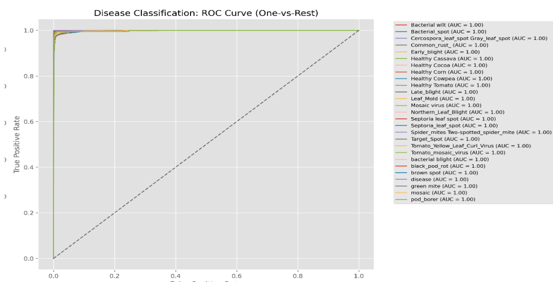
The second head, which had the slightly more intricate task of identifying diseases across 27 total classes (21 diseases + 5 healthy states + 1 OOD), performed exceptionally well:

Metric	Score
Overall Accuracy	98.20%

Macro F1-Score	0.9849
ROC AUC (OvR)	0.9996



Disease Confusion Matrix



Disease ROC curves and ROC AUC scores

The high obtained F1 score re-affirms the good performance of the model even in the case of minor diseases in the dataset. The perfect or near-perfect ROC AUC scores indicate excellent class separability in the current dataset. This is attributed to the distinct visual signatures of each disease and the effectiveness of the multi-task feature extraction, though real-world deployment may show slightly reduced metrics due to greater variability.

3. Per-Class Analysis

A more detailed look at the per-class performance shows excellent discrimination capabilities across both the plant identification head and disease identification head. As reflected in the confusion matrix (Figure 1), the plant head demonstrates near-perfect diagonal classification, with nearly all predictions classifying perfectly to the actual labels.

A multi-class ROC AUC of 1.00 indicates absolute separability of a positive class from the rest, as in the case of applying the One-vs-Rest approach. This implies that the model is effectively free from issues related to class discrimination within this dataset. The ROC curves (Figure 2) show optimal separability for all classes, which indeed serves as a good proxy for the robustness of feature extraction done by the model. This high performance is indicative of each class having highly distinctive visual features that the model can apply with absolute confidence when identifying class boundaries. This becomes all the more promising for applications concerning agriculture, where misclassifications may eventually have real-world consequences. The very diagonal matrix form of the confusion matrix also visually confirms that confusion between various classes is virtually non-existent for this plant identification problem. The head of this disease identification too enjoys the benefits of this clarity of classification.

Plant Class ROC AUC Scores

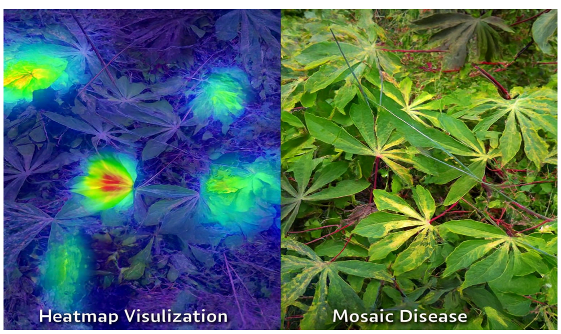
Plant Class	ROC AUC (OvR)
Cassava	1.00
Cocoa	1.00
Corn	1.00
Cowpea	1.00
Tomato	1.00
Unknown (OOD)	1.00

Disease Class	ROC AUC Scores
All 27 classes	0.9996

This illustrates the high confidence the model possesses, along with its ability to discriminate exceptionally well among all the defined classes in this specific dataset.

E. Interpretability Analysis using Grad-CAM

To further improve the transparency and reliability of the predictions provided by the model, we used Gradient-weighted Class Activation Map (Grad-CAM), which allowed us to view the parts of the input image that had the greatest impact on the decision produced by the network. The above technique also aided in the validation of whether the network is using relevant features or not, i.e., whether the focus was on the lesions, discolorations, and textural patterns, or not on the background or irrelevant features. Below, we have provided examples of Grad-CAM for a number of critical diseases and health classes on the two key crops, Corn, and Cassava.

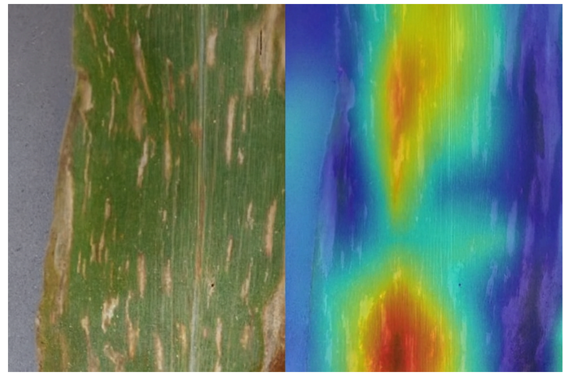


Cassava Mosaic Disease

Model highlights mosaic-like chlorotic patterns and leaf distortion, key visual cues for viral infection.

Healthy Cocoa

Broad, low-intensity activation across the pod and leaf surfaces, with no pathological focal points.



Corn Cercospora Leaf Spot

Activation centers on elongated, necrotic lesions with grayish centers, consistent with fungal leaf spot morphology.

These visualizations demonstrate our confidence in the alignment of the model's attention with the expected diagnostic regions by plant pathologists. The ability to explain predictions generated by the model is particularly beneficial in an agricultural context, in which farmers need diagnoses to be accompanied by justifications.

VI: CONCLUSION AND FUTURE DIRECTIONS

This paper's dual-head classifier serves as a foundation toward an automated and scalable agricultural diagnostic tool. Given that the current model produces very good accuracy for common species-pathogen identification, in moving from the laboratory prototype to a global agricultural solution, the expansion in scale has to be considerable. While the current model demonstrates exceptional performance on a curated dataset of 43,000 images, scaling to encompass global agricultural diversity represents the next frontier. Future work will expand systematically, considering several critical research trajectories.

A. Scaling with Large Biodiversity and Variety in Pathogens

The leap from a controlled dataset into one 1000 times its size introduces the "long-tail" problem commonly encountered in biological data. While our model currently handles popular diseases, as we move into hundreds of plant species, we will address the fact that some plants are prone to a myriad of unique pathogens.

Specialized disease branching will be done in future iterations of this work. The architecture will become a hierarchical one instead of having a generic "disease head." If the plant head detects a species with a high degree of confidence, the model will dynamically trigger sub-classifiers only trained on the 50-100 diseases affecting that particular genus. This avoids the "dilution" of accuracy that occurs when a model is forced to choose between thousands of irrelevant disease labels simultaneously.

B. Implementation of Biological Constraints and Logical Filtering

One of the biggest challenges with any classification problem on a large scale lies in the presence of "false positives" that are biologically nonsensical or impossible, for instance, a plant that is labeled as being "Late Blight" when the plant in question is really a succulent.

In order to address this, we also recommend the incorporation of a Biological Constraint Layer. Here, a hard matrix of ontological constraints is incorporated during the post-processing phase of the neural network. In this case, if a plant head detects a particular type of plant as "Plant X," a logical constraint will be activated to permit only diseases it is known to infect. This would be referred to as a "Knowledge-Informed Machine Learning" (KIML), and it will greatly help to reduce the overall level of noise within our large datasets.

C. Transition to Multi-Input, Multi-Output (MIMO)

It has been observed that a plant's health is not something that can be observed or determined visually, as its health is dictated by its surroundings. In order to attain professional-grade accuracy, the next step the author wishes to take with this research will be to shift away from image classification alone and embrace a classification known as Multi-Input, Multi-Output, or MIMO.

By adding another layer of metadata in the form of vectors around factors such as temperature, humidity, pH level of the soil, and geographical position, the model is able to "verify" the visual symptoms. This is because, for example, fungi have ideal conditions to thrive in. Therefore, taking a visual identification of a fungus into consideration and adding another layer in which the humidity level is described as 0% in the desert, the above model would be more robust because it resembles a real-life diagnostic process performed by a human agronomist.

D. From Classification to Prescription: Confidence-Based Therapeutics

The final aim of this particular research is not just to name the disease but to cure it too. We intend to fill the gap that is existing presently between computer vision and agricultural intervention by means of the 'Disease Confidence Scores'.

In our new framework, the softmax output from the disease head will be used for something other than a label. It will also be used as a "severity and certainty metric." We propose the implementation of a prescription engine, which maps the score to specific actions in the agricultural field:

- Low Confidence: The system prompts for a second image or a manual leaf swab.
- High Confidence/Low Severity: The system may propose organic or preventative options.
- High Confidence/High Severity: The system might develop a specific intervention plan (i.e., it might recommend the kind of fungicide used, among others).

This transformation changes the role from the classifier being simply a passive observer, meant for potential use as a tool for crop management, rather than as mere intelligence.

E. Final Summary

In conclusion, the research has shown that the dual-head architecture, where the classification of the plant and the diagnosis of the disease are done separately, is a highly effective approach to the creation of a classifier that scales.

Although this study has a controlled dataset, the structure has been built to perform better with even more data, rising to as high as 1,000 times its present amount. At that point, the model will be able to:

1. Apply Biological Logic: "Rules" to make sure the computers will only predict diseases that are really possible to be found on that particular plant.
2. Incorporate Environment Data: Utilize a multi-sensor which includes humidity and temperature, to make sure if the environment is matching a suspected disease or not.
3. Provide Treatments: Utilize "confidence scores" to advance a diagnosis to a degree where specific "cures" or interventions can be proposed.

Finally, this framework opens the door to eventually championing a universal diagnostic tool. In other words, the framework is designed to evolve in line with worldwide agricultural data to ensure farmers protect their food produce.

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